How Movie Reviewers Change Their Minds

Section: Taste Evolution in Amazon Movie Reviews Network

Megan Millet

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The Question

Do people's movie preferences change over time?

What We Know

Online rating systems capture billions of reviews, but we rarely ask: *Do individual reviewers become more critical or more generous as they mature on a platform?*

Why It Matters

- Recommendation algorithms assume stable preferences
- Understanding taste drift improves personalization
- Reveals how users learn to evaluate content

Dataset & Approach

Amazon Movie Reviews

- 1997–2012 (15 years)
- 5% random sample
- ∼100,000 reviews
- 66,556 reviewers
- 25,707 movies

Network Model

- Bipartite graph
- Reviewers ↔ Movies
- Edge = review event
- Rating: 1–5 stars

Measuring Taste Change

For each reviewer active across multiple years:

$$\Delta_i = \bar{r}_i^{\text{last}} - \bar{r}_i^{\text{first}}$$

where $\bar{r}_i^{\text{first}}$ and $\bar{r}_i^{\text{last}}$ are mean ratings in first and last active years.

Three Types of Reviewers

Among highly active users (50+ reviews):

Critics

Average rating ≤ 3.5 stars

Consistently harsh evaluators, 14.9% of users

Neutral

Average rating 3.5–4.0 stars

Made up 40.4% of users

Enthusiasts

Average rating ≥ 4.0 stars

Consistently generous, 44.7% of users

Key Insight

Users don't distribute evenly: rating behavior is **heterogeneous** and **stable within individuals**.

Main Finding: Reviewers Become Less Critical

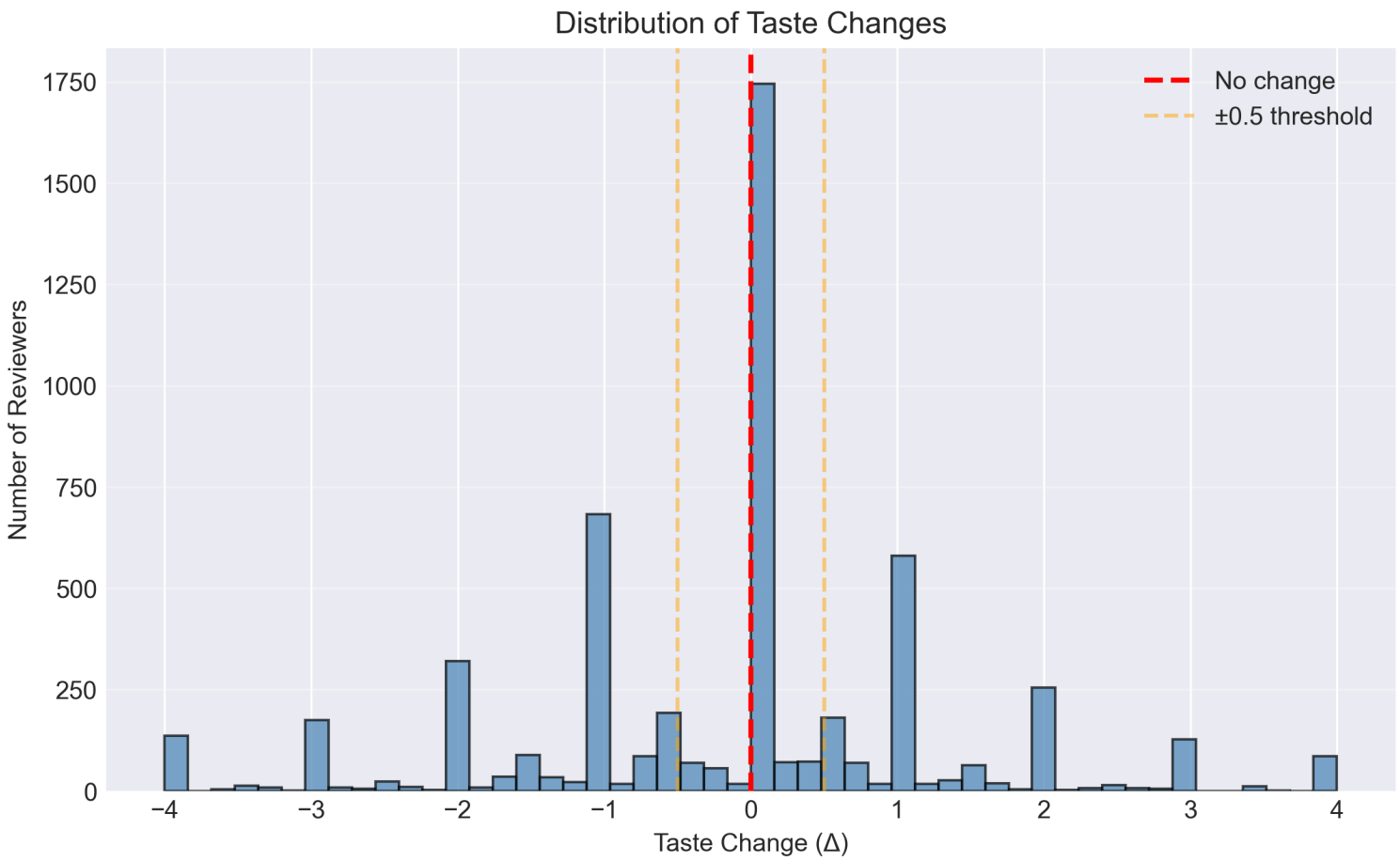
Among multi-year users:

- Significant taste changes observed ($|\Delta_i| > 0.5$ stars)
- **Majority became less critical** (higher ratings over time)
- Changes were **gradual**, not abrupt
- Some critics $\rightarrow$ neutral or enthusiasts

Possible Explanations

1. **Learning effect**: Users get better at finding movies they'll enjoy
2. **Platform maturation**: Improved recommendations help users avoid bad movies
3. **Selection bias**: Users increasingly review movies they expect to like
4. **Genre drift**: Shifting from critical genres (drama) to enjoyable ones (action/comedy)

Taste Change Distribution



Why This Matters

For Recommendation Systems

- Current systems assume **static preferences**
- Our findings: preferences **evolve systematically**
- Opportunity: Weigh recent reviews more heavily
- Opportunity: Model future taste trajectories, not snapshots

For Platform Design

- Early reviews may not represent long-term user satisfaction
- Users become more sophisticated over time
- Consider "user age" when aggregating ratings

Broader Impact

Understanding taste evolution helps platforms balance *exploration* (discovering new preferences) and *exploitation* (serving known preferences).

Conclusions

What I Found

- Reviewers' tastes **are not static**
- Majority drift toward **higher ratings** over time
- User types (critic/neutral/enthusiast) provide interpretable structure
- Changes are **gradual and measurable**

Next Steps

- Do taste-changers migrate between communities?
- Can we predict future taste drift from early reviews?
- How does taste change correlate with genre preferences?

Thank you!

Questions?